Poker Exercises 1

- A poker hand contains 5 out of 52 cards. The cards are 13 kinds (1-Ace, 2,..,10,J,Q,K) and 4 suites (spades, hearts, clubs, diamonds)
- What is the total number of poker hands?
- C(52,5)
- How many hands contain 5 different kinds?
- There are C(13,5) ways to choose the 5 kinds. For each card there are 4 choices of suit: $4^5 \cdot C(13,5)$
- How many hands contain the Ace of Spades and the King of Spades?
- Since these 2 cards are fixed, I have C(50,3) choices for the remaining 3 cards.
- How many hands contain exactly 1 Ace?
- There are 4 ways to choose the Ace. Once the Ace is selected, there are C(48,4) ways to choose the remaining four cards: $4 \cdot C(48,4)$